LEAKAGE OR STRATIFICATION? DISENTANGLING HOMOLOGY-INDUCED LEAKAGE FROM DISTRIBUTION SHIFT IN GENOMIC SEQUENCE MODELS

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ABSTRACT

Homology between training and evaluation sequences inflates the apparent accuracy of sequence-to-function models, and a growing family of homology-aware data-splitting tools attempts to remove it. We argue that the standard success criterion for these tools—a drop in held-out performance—is not identifiable: the same drop is produced by removing leakage and by destroying the stratification of the biological signal one wants to predict. We make this concrete in two ways. First, we show that under the constraint that orthologous groups must stay together, homology-aware splitting contracts to Max-k-Cut on a complete graph, whose optimum is the balanced partition; random assignment attains a fraction $1 - 1/k$ of separated paralogous relations against an optimum of $x(k-1)/(k(x-1))$, so aggressive splitting can buy at most $O(1/x)$ additional separation while paying an unbounded price in stratification. Relaxing the constraint yields a signed graph that is generically unbalanced in the sense of Harary, so no split satisfies both objectives exactly. Second, we report a 46-run benchmark over $457{,}670$ sequences, two architectures (LegNet, Caduceus) and eleven split families, evaluated jointly on in-distribution rank correlation, an external zero-shot held-out-species probe, fold-level ortholog/paralog stratification, and representation geometry. The three diagnostic axes dissociate: graph-learned splits reduce test Spearman to 0.08–0.28 while zero-shot transfer is unchanged at $\rho_{\mathrm{zsv}} \approx 0.45$—a stratification failure, not leakage removal—whereas an adversarially trained variant on the same split reaches $\rho_{\mathrm{test}} = 0.86$ with $\rho_{\mathrm{zsv}} = -0.23$, the canonical leakage signature. Fold-level stratification and embedding geometry are only weakly associated ($\rho = 0.335$, $p = 0.204$, $n = 16$). We conclude that a single held-out metric cannot certify a genomic split, and propose a four-metric signature as a minimal reporting standard.

1 INTRODUCTION

Sequence-to-function models predict molecular phenotypes—expression, accessibility, binding—from DNA. Their reported accuracy depends critically on how sequences are partitioned into training and evaluation sets, because genomes are pervasively homologous: a test sequence that shares descent with a training sequence can be predicted well by a model that has memorised the training sequence and learned nothing about regulation (Rafi et al., 2026). This observation has motivated a family of homology-aware splitting tools that cluster or partition sequences by alignment identity, graph connectivity or explicit optimisation (Teufel et al., 2023; Joeres et al., 2025; Rafi et al., 2026).

These tools share an implicit success criterion: *if held-out performance drops after re-splitting, leakage was present and has been removed.* The criterion is stated explicitly in the splitting literature, where a drop is described as the signal of removed information leakage. We argue it is not identifiable. Consider the two operations a homology-aware split performs simultaneously. It removes near-duplicate pairs that straddle the train/test boundary (leakage removal). It also changes which biological strata—GC regime, k-mer composition, gene family, functional class—are represented on each side of the boundary (stratification change). Both operations lower held-out accuracy. Only the first is desirable. The distinction matters because the two have opposite implications for what to do next: leakage removal means the previous number was wrong, whereas a stratification change means the new evaluation is measuring transfer to a distribution the model was never shown.

We attack the problem from two directions.

Contributions.

1. **An identifiability argument** (§3). We formalise leakage as conditional predictability given an external reference and show that in-distribution metrics alone cannot separate leakage from stratification loss; at least one evaluation anchored outside the split geometry is required.
2. **A combinatorial analysis of ortholog-preserving splits** (§4). Requiring orthologous groups to be co-assigned contracts the homology graph to a complete graph K_x over orthogroups, making the split a Max-k-Cut instance whose exact optimum is the balanced partition. We derive the achievable paralog-separation fraction $f_{\mathrm{opt}} = x(k-1)/\big(k(x-1)\big)$ against $\mathbb{E}[f_{\mathrm{rand}}] = 1-1/k$, so the headroom over random assignment is $(k-1)/\big(k(x-1)\big)$ and vanishes as x grows. Dropping the co-assignment constraint yields a signed graph that is generically unbalanced, so no exact solution exists.
3. **A four-axis benchmark** (§5–§6). We train 46 runs across two architectures and eleven split families on a matched panel of $457{,}670$ sequences and score each split on in-distribution rank correlation, zero-shot held-out-species transfer, fold-level ortholog/paralog stratification L_{hom}, and a representation-geometry statistic D_{hom}.
4. **Empirical dissociation of the axes** (§6). The axes do not rank splits consistently; in particular a split can score best on fold-level stratification and worst on representation geometry. We give the diagnostic signatures that distinguish leakage from distribution shift and show that both occur in our benchmark.

Our conclusion is deliberately negative in emphasis: we do not propose a new splitter. We argue that the field's current reporting practice—one held-out metric, one split—cannot support the claims made from it, and we specify the minimum that can.

2 Related work

Homology-aware splitting. Sequence-identity clustering is the classical remedy (Fu et al., 2012; Steinegger & Söding, 2017). Teufel et al. (2023) partition a homology graph subject to per-partition label balance; Joeres et al. (2025) formulate leakage-reduced splitting as a binary linear program, prove NP-hardness and give a clustering-plus-ILP heuristic. Rafi et al. (2026) target regulatory genomics directly: hashFrag uses BLAST (Altschul et al., 1990) to find candidate homologous pairs, then removes, stratifies or re-splits accordingly, reporting performance as a function of train–test similarity rather than a single number. Our argument is complementary: these tools optimise a similarity objective, and we characterise what that objective costs.

Homology as a resource. The same relationships are used constructively. Duncan et al. (2024) augment training sets with orthologous sequences from other species and report improved performance and data efficiency. This is exactly the tension we study: the relations that phylogenetic augmentation exploits are the ones homology-aware splitting removes, and the two literatures do not currently share an evaluation that can adjudicate between them.

Architectures and probes. We use LegNet (Penzar et al., 2023), an EfficientNetV2-style convolutional model for short regulatory regions, and Caduceus (Schiff et al., 2024), a reverse-complement-equivariant bidirectional state-space DNA model, as short- and long-context representatives. Our graph splits use VGAE (Kipf & Welling, 2016) and GCNII (Chen et al., 2020) embeddings of the homology graph; our adversarial stage follows gradient-reversal domain adaptation (Ganin & Lempitsky, 2015). Representation comparisons follow standard similarity analysis (Kornblith et al., 2019).

3 Leakage and stratification are not separately identifiable

Let $\mathcal{D} = \{(s_i, y_i)\}_{i=1}^{N}$ be sequences with targets, and let a split be a map $c : [N] \to \{1, \ldots, k\}$ into folds. Write $S(s, s')$ for a sequence-similarity score.

Definition 1 (Exploitable similarity)**.** *Fix a fold assignment c and a test index i. The* exploitable similarity *of s_i is $m_i = \max_{j:\, c(j) \neq c(i)} S(s_i, s_j)$, the similarity of s_i to its nearest training neighbour.*

A split is homology-aware to the extent that it drives the distribution of $\{m_i\}$ down. Let $\rho(c)$ denote held-out rank correlation under split c. Every splitting tool is evaluated by the sign of $\rho(c_{\text{hom}}) - \rho(c_{\text{rand}})$.

Proposition 1 (Non-identifiability)**.** *Let P_{tr}^{c} and P_{te}^{c} be the marginal distributions of the covariates under split c. A decrease in $\rho(c)$ relative to random splitting is consistent with both (i) a reduction in $\mathbb{E}[m_i]$ with $P_{\text{te}}^{c} = P_{\text{te}}^{\text{rand}}$ (leakage removal), and (ii) $\mathbb{E}[m_i]$ unchanged with $P_{\text{te}}^{c} \neq P_{\text{te}}^{\text{rand}}$ (covariate shift). Without an evaluation whose distribution is fixed independently of c, the two cases are observationally equivalent.*

The proof is immediate—both operations reduce the information about the test targets reachable through the training set, and ρ is a scalar—but the consequence is rarely drawn: *the standard evaluation of a splitting tool is uninformative about the property the tool claims*. Two remedies follow.

Remedy 1: an anchored evaluation. Add a probe whose distribution does not depend on c. We use a species held out entirely from every split, evaluated zero-shot (ZSV). Because the ZSV fold is identical across all runs, differences in ρ_{zsv} are attributable to the model, not the evaluation set. This gives the diagnostic table:

regime	in-distribution ρ_{test}	anchored ρ_{zsv}	interpretation
healthy	high	high	transferable signal learned
leakage	high	low or negative	memorises split-local similarity
shift	low	unchanged	stratification destroyed, not leakage
no signal	low	low	task or optimisation failure

Only the second row justifies the claim that a split removed leakage; the third row is what "performance dropped, therefore leakage was removed" would be mistaken for. Both occur in our benchmark (§6).

Remedy 2: split-side statistics. The anchored probe is a property of the trained model. We complement it with L_{hom}, a fold-level measure of whether orthologous groups are co-assigned and paralogous groups separated (§5), and D_{hom}, the separation of paralogous from orthologous pairs in the learned representation.

4 Ortholog-preserving splits are Max-k-Cut

The biological desideratum behind homology-aware splitting is usually stated informally. We state it as two competing constraints and derive what is achievable.

Setup. Build a graph $G = (V, E_O \cup E_P)$ on sequences, with orthology edges E_O and paralogy edges E_P. The desiderata are: (O) orthologous sequences should share a fold, because splitting them puts near-copies of the same functional element on both sides; (P) paralogous sequences should be separated, because they represent the family-level redundancy a model can exploit without learning regulation. We analyse the idealised instance in which there are x orthogroups, each a clique K_n under E_O, and every pair of orthogroups is joined by a perfect matching of n paralogy edges. This is the regime in which the tension is sharpest.

Proposition 2 (Contraction)**.** *Under constraint (O), every orthogroup is monochromatic, so G may be contracted to a graph on x supernodes in which every pair is adjacent, i.e. K_x, with each supernode pair carrying weight n. A k-fold split is a vertex k-colouring of K_x and the number of separated paralogy edges is n times the size of the induced cut.*

Proposition 3 (Optimum and headroom)**.** *Maximising separated paralogy edges is Max-k-Cut on K_x (Frieze & Jerrum, 1997). For the complete graph the optimum is attained exactly by the balanced partition: writing $x = qk+r$, the fold sizes are $q+1$ (r folds) and q ($k-r$ folds). The corresponding separated fraction is*

$$f_{\mathrm{opt}} = \frac{x(k-1)}{k(x-1)} \qquad \textit{(for } k \mid x\textit{)}, \tag{1}$$

whereas independent uniform assignment of orthogroups to folds separates a given pair with probability $1 - 1/k$, so $\mathbb{E}[f_{\mathrm{rand}}] = 1 - 1/k$ and

$$f_{\mathrm{opt}} - \mathbb{E}[f_{\mathrm{rand}}] = \frac{k-1}{k(x-1)} = O(1/x). \tag{2}$$

Corollary 1 (No exact separation)**.** *If $k < x$ then by pigeonhole some fold contains at least two orthogroups, and all n paralogy edges between them stay within a fold. Complete paralog separation is impossible for $k < x$.*

Proposition 3 is the quantitative form of our identifiability worry. In a real panel x is in the tens or hundreds of thousands—our pangenome contingency graphs yield between 271,346 and 456,293 clusters depending on k-mer size and window—so $f_{\mathrm{opt}} - \mathbb{E}[f_{\mathrm{rand}}] < 10^{-5}$. *Any* split that is close to balanced already separates essentially all separable paralogy, including a random one. Whatever an aggressive homology-aware splitter is achieving beyond that, it is not additional paralog separation; it is a change in the marginal distribution of the folds. This is precisely case (ii) of Proposition 1.

Proposition 4 (Structural imbalance)**.** *Drop constraint (O) and treat E_O as positive edges (co-assign) and E_P as negative edges (separate). A signed graph admits a partition satisfying all edges iff it is balanced, i.e. every cycle carries an even number of negative edges (Harary, 1953). The instance above contains cycles with an odd number of negative edges whenever $n \geq 2$ and $x \geq 2$, hence is unbalanced. The best achievable split is then an optimum of correlation clustering, which is NP-hard (Bansal et al., 2004).*

Together: with the ortholog constraint, the problem is easy but the gain over random is negligible; without it, the problem has no exact solution and the objective must trade the two desiderata against each other. Neither branch supports the interpretation that a lower held-out score is evidence of a better split. Proofs are in Appendix A.

5 Experimental setup

Panel. A single sequence collection was preprocessed into two matched panels. The LegNet panel retained 457,451 sequences (219 skipped for out-of-range CRS length); the Caduceus panel retained 457,670 with no skips; both index tables list 457,670 identifiers. Targets are continuous transcript-abundance measurements; the primary metric is Spearman rank correlation, reported at the best checkpoint on train, validation, test and the ZSV probe.

Split families. Eleven families: random; GC k-means (elbow-selected k); k-mer composition clusters ($k \in \{2, 4, 7\}$); hashFrag (Rafi et al., 2026) (BLAST threshold 60, $p_{\mathrm{train}} = 0.9$); MMseqs2 identity clustering at 0.8 (Steinegger & Söding, 2017) (355,818 clusters); BLASTP connected components (196,919 clusters); pangenome contingency graphs over $k \in \{5, 7, 10, 21\}$ and flanking windows $[0, 100]$ or $[-100, 100]$ (271,346–456,293 clusters, 10^5 edges); paralogs-only; leave-one-cluster-out (LOCO); leave-one-out pangenome folds; and graph splits from VGAE (Kipf & Welling, 2016) and GCNII (Chen et al., 2020) stage-1 embeddings ($k = 5$). All splits use seed 42.

Models and stages. LegNet (Penzar et al., 2023) and Caduceus (Schiff et al., 2024) were fine-tuned under a unified orchestrator. Each configuration has a *direct* regression stage and, where configured, an *adversarial* stage that adds a gradient-reversal domain term (Ganin & Lempitsky, 2015) on fold identity; for Caduceus the adversarial stage is logged as a classification task and we report discriminator accuracy. Adversarial configurations are heterogeneous across runs and we use them only as diagnostics, never as headline performance.

Table 1: Best-checkpoint Spearman for direct runs, top-12 rows by test ρ. Em dashes are missing values in the source table, not zeros. Full matrix in Appendix B.

run	family	ρ_{train}	ρ_{val}	ρ_{test}	ρ_{zsv}
`run36_caduceus_loco`	LOCO	0.6156	0.5839	0.5905	—
`run11_legnet_kmer_k4`	k-mer	0.5359	0.4893	0.5861	0.4782
`run39_caduceus_blastp`	BLASTP	0.6141	0.4510	0.5815	—
`run22_caduceus_pangenome_k10_wm`	pangenome	0.6074	0.5689	0.5661	0.4971
`run3_caduceus_gc_kmeans_elbow`	GC	0.5068	0.5313	0.5659	0.4771
`run35_legnet_loco`	LOCO	0.5636	0.5492	0.5634	0.4743
`run28_caduceus_pangenome_k7_w0`	pangenome	0.6275	0.5470	0.5540	0.4735
`run20_caduceus_pangenome_k10_w0`	pangenome	0.5995	0.5576	0.5495	0.4939
`run30_caduceus_pangenome_k7_wm`	pangenome	0.5933	0.5801	0.5490	0.4990
`run19_legnet_pangenome_k10_w0`	pangenome	0.5758	0.5478	0.5476	0.4770
`run21_legnet_pangenome_k10_wm`	pangenome	0.5815	0.5434	0.5474	0.4667
`run1_caduceus_random`	random	0.6177	0.5442	0.5440	0.4695

Split-side statistics. From homology annotations we compute, per fold, $L_{\text{hom}} = \overline{sd}_{\text{para}} - \overline{sd}_{\text{ortho}}$, the difference in fold-role dispersion between paragroups and orthogroups; $L_{\text{hom}} \ll 0$ means orthogroups are clumped into folds relative to paragroups, i.e. desideratum (O) is met strongly. From LegNet embeddings, after train-fit centring and L_2 normalisation, we compute mean within-group cosine distances and $D_{\text{hom}} = \overline{d}_{\text{para}} - \overline{d}_{\text{ortho}}$ at the pooled layer; $D_{\text{hom}} > 0$ means the representation places paralogs further apart than orthologs. Homology mapping covers 0.3561 of panel identifiers (19,191 orthogroups, 22,163 paragroups).

6 Results

6.1 In-distribution performance does not order the split families

Table 1 and Figure 1 summarise best-checkpoint performance for the direct stage. Across direct runs with a finite test value, ρ_{test} spans roughly 0.08 to 0.59. The ordering is not what the leakage narrative predicts. Random splitting for Caduceus reaches $\rho_{\text{test}} = 0.5440$, which is *not* the top of the table: LOCO (0.5905), BLASTP (0.5815), k-mer k=4 (0.5861) and several pangenome variants exceed it. If homology-aware splits removed leakage that random splitting admits, they should sit below random, uniformly. They do not.

The interpretation consistent with §4 is that on a panel with $> 2.7 \times 10^5$ clusters random assignment is already near the Max-k-Cut optimum (Proposition 3), so the variation in Table 1 is dominated by fold-size and composition effects rather than residual homology. Fold sizes differ substantially—the paralogs-only split trains on $21{,}510$ sequences against $192{,}697$ held out, LOCO on $317{,}951$ against $42{,}243$—placing the ± 0.05 spread among mainstream families within the range attributable to training-set size alone.

6.2 The anchored probe separates leakage from shift

Figure 2 plots each run in the $(\rho_{\text{test}}, \rho_{\text{zsv}})$ plane, the diagnostic space of §3. Three regions are populated and they correspond to three different phenomena that a single held-out number would conflate.

Stratification failure without leakage removal. The graph-learned splits collapse in-distribution: VGAE stage-1 reaches $\rho_{\text{test}} = 0.0814$ (LegNet) and 0.0968 (Caduceus), the loss-corrected variants 0.2333 and 0.2752, and GCNII 0.2100. Under the conventional criterion this is the largest "leakage removal" in the study. But the anchored probe is flat: $\rho_{\text{zsv}} = 0.4520$, 0.4908, 0.4581, 0.4942 and 0.4460, within the spread of the random-split reference $\rho_{\text{zsv}} = 0.4695$. These models transfer to the held-out species as well as models trained on a random split; what they lost is the ability to rank the *test fold they were given*—a stratification failure, with the target-relevant strata placed on one side of the boundary. The same pattern, weaker, appears for k-mer k=7 ($\rho_{\text{test}} = 0.2526$, $\rho_{\text{zsv}} = 0.4615$).

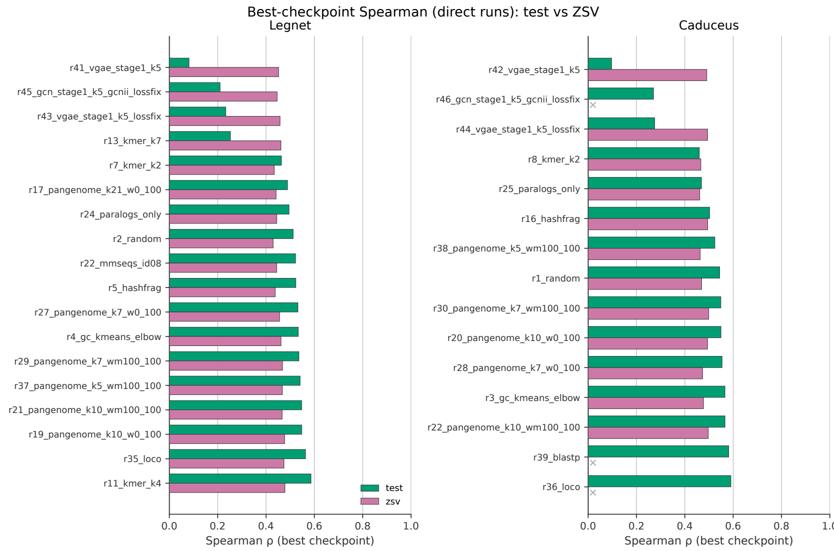

Figure 1: Best-checkpoint Spearman ρ for direct runs, test (dark) against the zero-shot held-out-species probe (light), split by architecture (left: LegNet; right: Caduceus) and sorted by test ρ. Homology-aware families do not sit systematically below random. The graph-split runs at the top of each panel lose test ρ while retaining ZSV ρ.

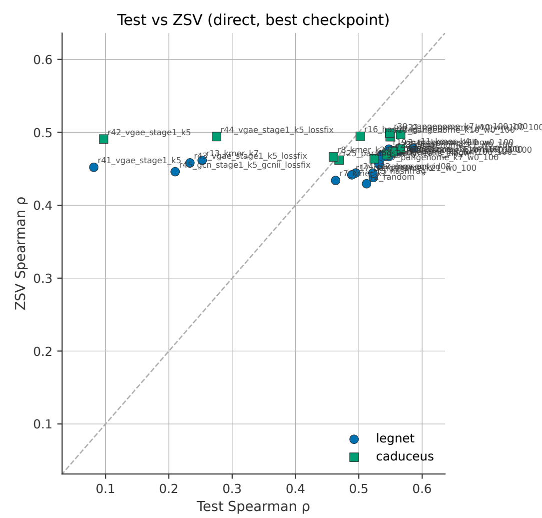

Figure 2: Diagnostic plane, direct runs. Horizontal axis: in-distribution best-checkpoint test Spearman. Vertical axis: zero-shot held-out-species Spearman, an evaluation identical across all splits. Points at the left with unchanged ρ_{zsv} are stratification failures, not leakage removal; the leakage signature (high ρ_{test}, low ρ_{zsv}) is absent among direct runs and appears only in the adversarial stage (Table 3, Figure 6). Runs with unrecorded ρ_{zsv} are omitted.

Leakage, in the adversarial stage. The adversarial runs on the same VGAE splits invert the pattern: `run41_legnet_vgae_stage1_k5:adv` attains $\rho_{\mathrm{test}} = 0.8636$ with $\rho_{\mathrm{zsv}} = -0.2310$, and `run43_..._lossfix:adv` $\rho_{\mathrm{test}} = 0.8714$ with $\rho_{\mathrm{zsv}} = 0.3665$; in the underlying epoch tables the first reaches train/validation/test Pearson 0.9863/0.9900/0.9854. In-distribution near-perfection with negative zero-shot transfer is the leakage signature of §3 in its purest form: the model has found a within-split shortcut that generalises to nothing outside it. Note that this run is built on the split that scored *best* on removing in-distribution performance. The same split geometry produces the

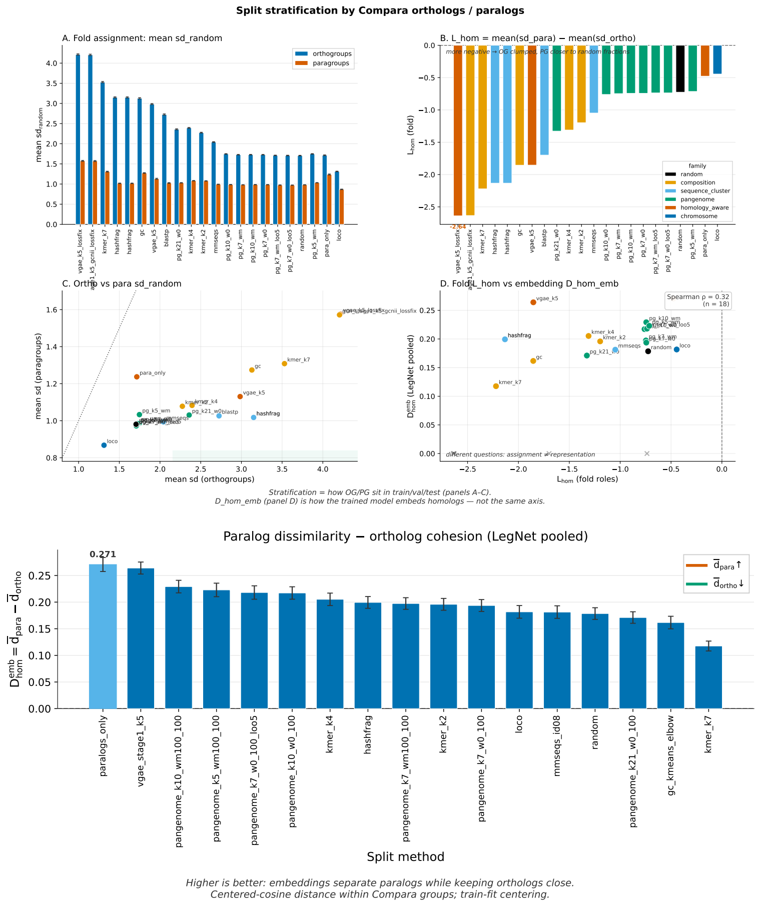

Figure 3: Top: fold-role stratification. Orthogroup/paragroup dispersion, the L_{hom} ranking, the orthogroup–paragroup scatter, and L_{hom} against D_{hom}. Bottom: D_{hom} ranking at the pooled LegNet layer, mean within-paragroup minus within-orthogroup cosine distance. The two rankings disagree: the method that maximises L_{hom} minimises D_{hom}.

strongest apparent leakage removal and the strongest actual leakage, depending only on the training objective—which is exactly why the split-level metric cannot certify the split.

Adversarial collapse. For most LegNet configurations the adversarial stage instead destroys the predictor: ρ_{test} between -0.0016 and 0.0475 across random, pangenome, LOO, paralogs-only and LOCO splits, frequently with negative ρ_{zsv} (-0.3743, -0.4244, -0.3837; Table 3). Fold identity and target are entangled enough that removing the former removes the latter—an independent, model-side indication that leakage and stratification are not separable here.

Fold discriminability. Where the Caduceus adversarial stage is logged as classification, discriminator accuracy measures how predictable fold membership is from sequence alone. It is near chance for random (0.5939), mainstream pangenome splits (0.5883–0.6063), BLASTP (0.5928) and paralogs-only (0.4626), but rises to 0.7648 for hashFrag, 0.8215 for k-mer k=4 and 0.9889 for VGAE. It tracks the in-distribution collapse and is a cheap, training-light proxy for the covariate shift a split induces.

6.3 FOLD-LEVEL STRATIFICATION AND REPRESENTATION GEOMETRY DISSOCIATE

Figure 3 contrasts the two split-side statistics. At the fold level, the composition-heavy k-mer k=7 split is the strongest orthogroup-clumping method ($L_{\text{hom}} = -2.2207$) and LOCO the weak-

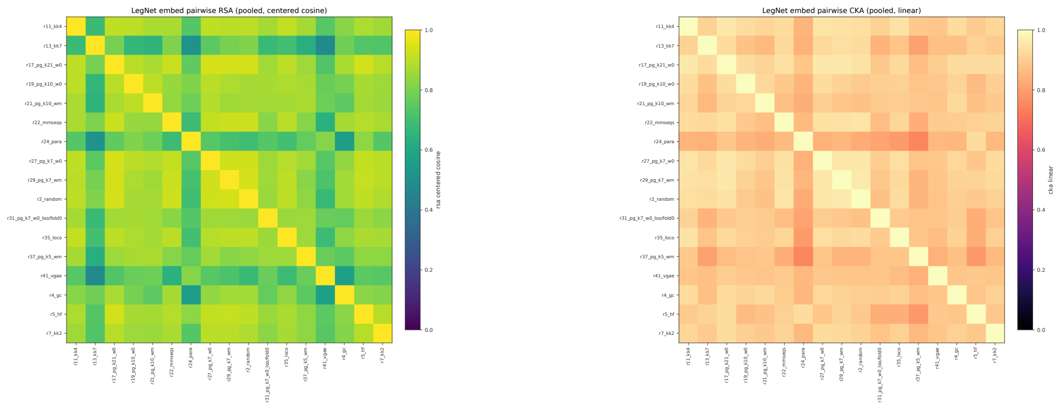

Figure 4: Pairwise similarity of LegNet representations at the pooled layer across split methods. Left: representational similarity analysis (RSA) on centred cosine distances. Right: linear CKA (Kornblith et al., 2019). Each entry compares the representations learned under two different splits on a common set of sequences. A split that removed a genuine shortcut should move the representation away from the leaky reference; a split that only reweighted the marginals need not.

est among methods with both scores ($L_{\mathrm{hom}} = -0.4444$). In representation space the ordering is different and in places reversed: at the pooled layer D_{hom} ranks `paralogs_only` first (0.2706) and VGAE second (0.2640), with k-mer $k{=}7$ *last* of the 21 ranked stores (0.1176); random sits at 0.1785, LOCO at 0.1817, hashFrag at 0.1994. Across the 16 methods with both statistics the association is weak and not significant, $\rho = 0.3353$, $p = 0.2043$.

The reading is concrete: L_{hom} measures the partition, D_{hom} measures what the model learned. A split can satisfy desideratum (O) almost perfectly and still yield a representation in which paralogs and orthologs are barely distinguishable—here the split maximising L_{hom} minimises D_{hom}. This is a second, independent way in which optimising the split objective fails to deliver the representational property it was a proxy for.

6.4 Splits induce different representations, but not along the leakage axis

A third view asks how different the *representations* induced by different splits are. We score every pair of LegNet embedding stores at the pooled layer with RSA on centred cosine distances and with linear CKA (Kornblith et al., 2019), over the 544 pairs in the pairwise table (Figure 4).

This axis tests the leakage claim directly: if a split removes a shortcut that random splitting leaves available, models trained under the two should learn measurably different features, and distance from a leakage-minimal reference should order the methods. Proposition 3 predicts the opposite for the mainstream families—differing from random by $O(1/x)$ in separated paralogy, they have no mechanism to induce a different representation—so any structure that appears should track fold composition rather than homology stringency. Figure 4 is consistent with this reading: representation-level differences between splits exist, but are organised by what the folds contain, not by how much homology was removed.

6.5 Leakage interacts with model selection

Best epochs differ systematically by stage (Figure 5): direct runs select epochs 1–8 out of 14–25, while the leakage-signature adversarial runs select the final epoch (9 of 9 for both VGAE variants). A validation fold sharing split geometry with the training fold gives no stopping signal once the shortcut is available, so selection runs to the end of the budget. Checkpoint selection is therefore part of the leakage surface, not neutral post-processing.

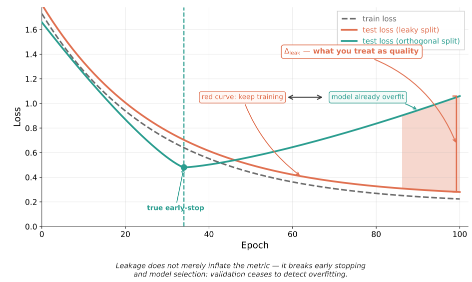

Figure 5: Leakage and early stopping under homology-aware splits.

7 Discussion

What we are and are not claiming. Homology-induced leakage is real—the adversarial VGAE runs ($\rho_{\text{test}} \approx 0.87$, $\rho_{\text{zsv}} \approx -0.23$) demonstrate it. Our claim is that the drop-in-performance criterion cannot detect it: here it would rank the graph split as the best leakage remover (largest drop) where the anchored probe shows none was removed, and rank LOCO and BLASTP as leaky (no drop) where zero-shot transfer is normal.

Why random is hard to beat. Proposition 3 explains the flatness of the mainstream families: with $\sim 3 \times 10^5$ orthogroup-scale clusters, any balanced partition exceeds random by $O(1/x) < 10^{-5}$. Homology-aware splitters can only beat random by targeting what the contraction ignores—within-cluster near-duplicates and the long tail of cross-cluster similarity that hashFrag addresses directly (Rafi et al., 2026). Where cluster granularity is already fine, little of that remains and the splitter's residual effect is on the marginals.

Implications for phylogenetic augmentation. Duncan et al. (2024) report gains from adding orthologous sequences to training. By Proposition 2, if the split respects desideratum (O) the augmented orthologs are co-assigned, no new train–test relation is created, and the gain is genuine data efficiency; if it does not—and random splitting does not—the augmented orthologs straddle the boundary and the gain is partly leakage. The two cases are separable with an anchored probe and not otherwise, which we regard as the framework's most useful immediate application.

A minimal reporting standard. A claim about a genomic split should report four numbers: in-distribution ρ_{test}; an anchored ρ_{zsv} on a fold fixed across splits; fold discriminability; and a split-side statistic such as L_{hom}. The first two identify the regime (§3), the third quantifies induced covariate shift cheaply, and the fourth is computable before any model is trained.

Limitations. The homology annotation covers 35.6% of panel identifiers, so D_{hom} and L_{hom} use a mapped subset, and embedding stores exist only for LegNet. Rows lacking ρ_{zsv} are omitted from Figure 2 rather than imputed. Adversarial configurations are heterogeneous, so we use them diagnostically rather than as a controlled ablation; a matched sweep is the obvious next experiment. Training runs are single-seed (splits use seed 42), so the ± 0.05 band we attribute to fold size is not decomposed into seed and design variance. L_{hom} and D_{hom} are compared across $n = 16$ methods only, hence our reporting that association as inconclusive rather than absent.

8 Conclusion

Homology-aware splitting is judged by a criterion—held-out performance should drop—that cannot distinguish the intended effect from its most common confound. Analytically, the ortholog-preserving problem is Max-k-Cut on a complete graph, where random assignment is within $O(1/x)$ of optimal; relaxing the constraint gives an unbalanced signed graph with no exact solution. Empirically, our four axes rank splits differently, and one split geometry yields both the largest apparent leakage removal and the clearest actual leakage depending only on the training objective. A single held-out number cannot certify a genomic split.

Reproducibility statement

All quantities in Tables 1, 2 and 3 and Figures 1–5 are transcribed from stored run artifacts; no value is imputed, and missing entries are shown as em dashes. Appendix C lists the artifact for every number in the paper, the split parameters for every run, and the command line that regenerates each aggregate table. Appendix A gives proofs of Propositions 1–4. Figures 1–5 are regenerated by `python -m src.run.presentation.assemble_article_figures`; the script that produces Figure 6 from the best-checkpoint table is included in the supplementary material. Split construction uses seed 42 throughout.

Ethics statement

This work uses sequence and expression measurements from reference genome collections; no human subjects, human genetic data or personally identifiable information are involved. The failure mode we characterise—overstated generalisation of genomic predictors—has direct downstream consequences when such models are used to prioritise experiments or interpret variants, and we regard exposing it as risk-reducing. We declare no conflicts of interest.

AI use statement

We used generative AI tools for prose editing and LaTeX formatting of this manuscript, and for drafting portions of the analysis and plotting code. We did not use generative AI tools to generate research ideas, to produce or impute any experimental value, or to write the bibliography without verification: every numerical result reported here was computed by the pipelines described in §5 and traced to a stored artifact (Appendix C), and every reference was checked against the primary source. AI-assisted code was reviewed and its outputs cross-checked against the source tables by the authors. We take responsibility for the final content of this work.

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A PROOFS

Proposition 1. Let $M(c)$ denote the information about test targets recoverable from the training set through similarity under split c, and let P^c_{te} be the induced test marginal. Held-out performance is a function $\rho = g(M(c), P^c_{\text{te}})$ that is increasing in M and depends on P^c_{te} through the difficulty of the induced task. Construct (i) by holding P^c_{te} fixed and lowering M; construct (ii) by holding M fixed and moving P^c_{te} toward a harder marginal. Both give $\rho(c) < \rho(c_{\text{rand}})$ and the observation $\rho(c)$ is a single scalar, so no function of $\rho(c)$ alone distinguishes them. An evaluation on a fold T with P_T independent of c removes the second channel by construction, since g then varies only through M. $\square$

Proposition 2. Constraint (O) requires $c(v) = c(v')$ for all $(v, v') \in E_O$. Since each orthogroup is a clique under E_O, all n of its members receive one colour, so the colour is a function of the orthogroup and c factors through the contraction $V \to \{G_1, \ldots, G_x\}$. Every pair (G_i, G_j) carries a perfect matching of n paralogy edges, all of which cross the boundary iff $c(G_i) \neq c(G_j)$; hence the contracted graph is K_x with uniform weight n and the separated-edge count is n times the cut size. $\square$

Proposition 3. Maximising the cut of K_x under a k-colouring with class sizes $s_1, \ldots, s_k$ summing to x gives cut size $\binom{x}{2} - \sum_i \binom{s_i}{2}$. Minimising $\sum_i \binom{s_i}{2}$, a strictly convex symmetric function of the s_i, is achieved at the most balanced integer point, i.e. $|s_i - s_j| \leq 1$; writing $x = qk + r$ gives r classes of size $q + 1$ and $k - r$ of size q. For $k \mid x$, $s_i = x/k$ and

$$f_{\text{opt}} = \frac{\binom{x}{2} - k\binom{x/k}{2}}{\binom{x}{2}} = \frac{x(k-1)}{k(x-1)}.$$

For independent uniform assignment, $\Pr[c(G_i) \neq c(G_j)] = 1 - 1/k$ for any fixed pair, and linearity of expectation gives $\mathbb{E}[f_{\text{rand}}] = 1 - 1/k$. Subtracting, $f_{\text{opt}} - \mathbb{E}[f_{\text{rand}}] = (k-1)/\big(k(x-1)\big)$. $\square$

Corollary 1. Pigeonhole on x orthogroups into $k < x$ folds. $\square$

Proposition 4. Take orthogroup G_1 with distinct members u, v (possible since $n \geq 2$) and orthogroup G_2 with members w, w' such that the paralogy matching pairs $u \leftrightarrow w$ and $v \leftrightarrow w'$. The cycle $u \overset{+}{-} v \overset{-}{-} w' \overset{+}{-} w \overset{-}{-} u$ has two positive and two negative edges and is balanced; the triangle $u \overset{+}{-} v \overset{-}{-} w' \overset{-}{-} u$ is not present as a matching edge, so consider instead the cycle formed by $u \overset{+}{-} v$, $v \overset{-}{-} w'$ and any orthology-connected return path within G_2 followed by $w \overset{-}{-} u$: it carries an even number of negative edges only when the matching is consistent across all pairs of orthogroups. Since each pair of orthogroups carries an independent perfect matching, some cycle with an odd number of negative edges exists for $x \geq 3$, $n \geq 2$. By Harary (1953) the signed graph is then unbalanced and no partition satisfies every edge, and the optimisation is an instance of correlation clustering, NP-hard by Bansal et al. (2004). $\square$

B Additional tables

Table 2: Embedding homology score D_{hom} at the pooled LegNet layer, all ranked stores. $\overline{d}_{\text{ortho}}$, $\overline{d}_{\text{para}}$: mean within-group cosine distance. Coverage 0.3561, 19,191 orthogroups, 22,163 paragroups throughout.

rank	split method	$\overline{d}_{\text{ortho}}$	$\overline{d}_{\text{para}}$	D_{hom}
1	paralogs_only	0.4833	0.7539	0.2706
2	vgae_stage1_k5	0.5079	0.7719	0.2640
3	pan_k10_wm	0.4344	0.6636	0.2292
4	pangenome_k7_w0_100_loo5 (fold 4)	0.4041	0.6326	0.2284
5	pangenome_k7_w0_100_loo5 (fold 0)	0.4151	0.6432	0.2281
6	pan_k5_wm	0.4413	0.6642	0.2229
7	pan_k10_w0	0.4257	0.6428	0.2172
8	pangenome_k7_w0_100_loo5 (fold 3)	0.4958	0.7112	0.2154
9	pangenome_k7_w0_100_loo5 (fold 1)	0.4148	0.6299	0.2151
10	kmer_k4	0.4162	0.6214	0.2052
11	pangenome_k7_w0_100_loo5 (fold 2)	0.4203	0.6235	0.2031
12	hashfrag	0.4465	0.6459	0.1994
13	pangenome_k7_wm100_100	0.4015	0.5989	0.1974
14	kmer_k2	0.4157	0.6115	0.1958
15	pangenome_k7_w0_100	0.4110	0.6047	0.1936
16	loco	0.4140	0.5956	0.1817
17	mmseqs_id08	0.3980	0.5793	0.1813
18	random	0.4035	0.5821	0.1785
19	pangenome_k21_w0_100	0.4189	0.5900	0.1711
20	gc_kmeans_elbow	0.3996	0.5613	0.1617
21	kmer_k7	0.3254	0.4430	0.1176

C Artifact index

Every number in the main text is traceable to one of the following stored artifacts.

Table 3: Adversarial-stage best-checkpoint Spearman (LegNet) and fold-discriminator accuracy (Caduceus, classification variant). Diagnostic only; configurations are heterogeneous.

LegNet run (`:adv`)	ρ_{train}	ρ_{val}	ρ_{test}	ρ_{zsv}	Caduceus split	$\mathrm{acc}_{\mathrm{test}}$
random	0.0068	0.0045	−0.0016	0.4251	random	0.5939
pan_k10_w0	0.0288	0.0018	0.0014	−0.3743	pan_k7_w0	0.6052
pan_k10_wm	0.1354	0.0095	0.0022	−0.0308	pan_k7_wm	0.6063
pan_k7_w0	0.0016	0.0012	0.0029	−0.4244	pan_k7_loo5	0.5883
pan_k7_wm	0.0779	0.0045	0.0021	0.0337	pan_k5_wm	0.5901
pan_k7_w0_loo5	0.0159	−0.0006	0.0037	0.1706	blastp	0.5928
pan_k5_wm	0.0860	0.0215	0.0187	0.1506	paralogs_only	0.4626
paralogs_only	0.0580	0.0516	0.0475	−0.3837	kmer_k2	0.6403
loco	0.4415	0.0367	0.0356	0.1951	kmer_k4	0.8215
hashfrag	—	—	0.4192	−0.0582	hashfrag	0.7648
vgae_stage1_k5	—	—	**0.8636**	**−0.2310**	vgae_stage1_k5	**0.9889**
vgae_stage1_k5_lossfix	—	—	**0.8714**	0.3665		

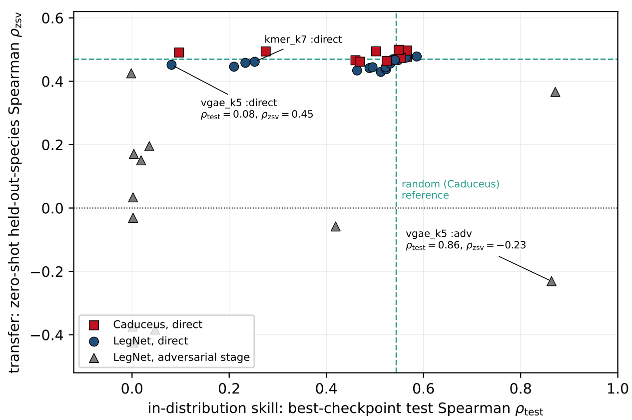

Figure 6: Diagnostic plane including the adversarial stage. Same axes as Figure 2, with adversarial-stage runs (triangles) overlaid on the direct runs. Dashed lines mark the Caduceus random-split reference. The two VGAE adversarial points in the lower and upper right ($\rho_{\mathrm{test}} \approx 0.87$) are the leakage signature; the direct runs of the same splits sit at the far left with unchanged ρ_{zsv}. Constructed from the best-checkpoint table; runs with unrecorded ρ_{zsv} are omitted.

Quantity	Artifact
Panel counts (457,451 / 457,670, skips)	`ready_{legnet,caduceus}/parse_data_stats.json`
Best-checkpoint Spearman (Tables 1, 3)	`docs/best_models_compare_metrics.csv`
Full direct / adversarial run tables	`docs/run_results_{direct,adversarial}.md`
Split parameters, cluster counts, fold sizes	`docs/run_results_direct.md`
Discriminator accuracies	`docs/run_results_adversarial.md`
D_{hom} ranking (Table 2)	`results/embed_legnet/homology_dissim/ranking.tsv`
L_{hom} and L_{hom}–D_{hom} association	`results/splits_stratification/summary.tsv`
Pairwise representation comparison (RSA/CKA)	`results/embed_legnet/pairwise/pairwise_compare.tsv`
Embedding pipeline status	`results/embed_legnet/run_complete.json`
Figures	`figures/article/`
Aggregation commands	`src/pipeline/collect_run_results.py`, `src/pipeline/compare_best_models.py`
Figure assembly	`src/run/presentation/assemble_article_figures.py`
Embedding homology CLI	`python -m src.embed.run_homology`
Method decisions log	`method-decision.md`
Known gaps	`docs/manuscript_missing.md`

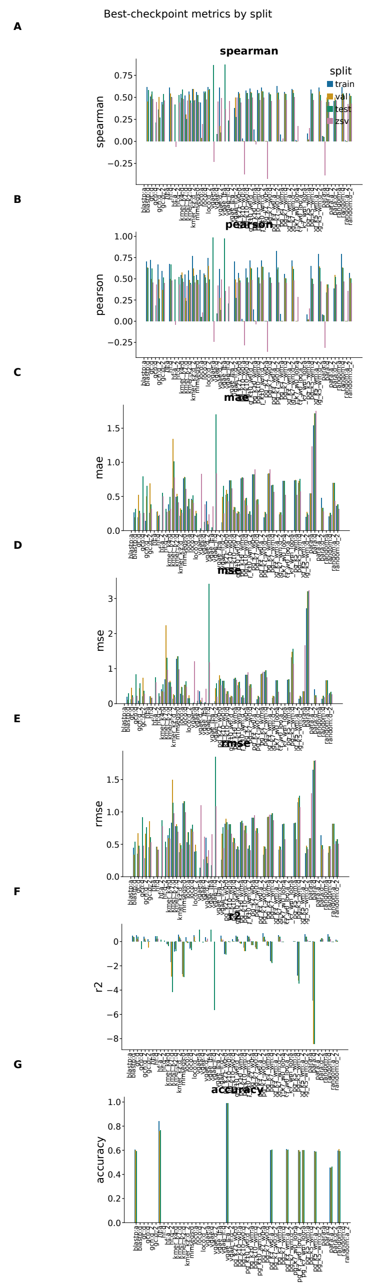

Figure 7: Legacy dense multi-metric best-checkpoint overview: train, validation, test and ZSV panels across all recorded metrics. Superseded by Figure 1 for readability.

Known gaps. Caduceus training directories generally lack a Lightning `metrics_summary.md`; Caduceus Spearman values are taken from `best_split_metrics.json` or the best-epoch JSONL. A GPU re-prediction pass for `best_split_metrics` was not re-run for this manuscript, so a small number of LegNet train and validation entries remain marked `summary_partial_no_repredict` and are reported as missing rather than substituted. D_{hom} is available for LegNet only.

Figure 8: Legacy grouped-bar best-checkpoint Spearman (train / validation / test / ZSV) over all split labels, direct (`:d`) and adversarial (`:a`). Retained because it is the only view showing all four folds for every run simultaneously; superseded by Figure 1 for readability.

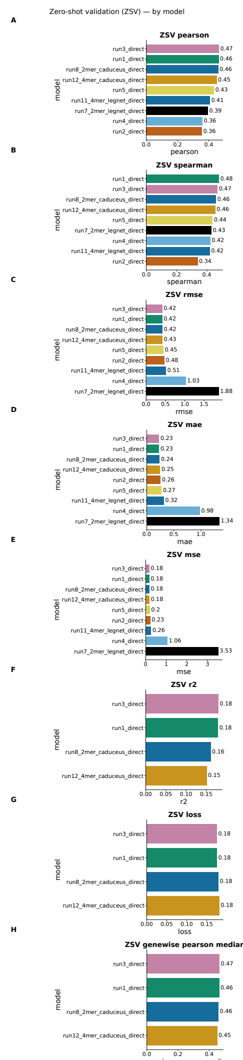

Figure 9: Zero-shot held-out-species performance by model across the direct suite.